

Gaurang Gupta

AI/ML Engineer | Data Science · Machine Learning · Predictive Analytics

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Education

Bennett University

2022 – 2026

B.Tech, Computer Science & Engineering (AI & ML Specialization) CGPA: 7.0/10

Experience

AI/ML Engineer Intern

Apr 2025 – Present

Tychi Labs

Delhi, India

- Developed machine learning based financial market prediction system leveraging time-series data and feature engineering techniques to generate trading signals with sub-2 second inference latency.
- Built AI-driven retrieval and ranking system utilizing vector embeddings, semantic similarity search, and performance evaluation methodologies to improve information retrieval accuracy.
- Designed and evaluated AI decision-making framework incorporating ensemble-style voting mechanisms and model output aggregation techniques.
- Built recurring-investment system on ERC-3009 permit-based USDC transfers, enabling gasless, signature-authorized automated crypto purchases
- Built Agent Execution Interface (AEI), an MCP-native gateway letting AI agents execute EVM transactions across 8 chains with no destination-chain gas — wallet-bound access keys, non-custodial signing, and x402 payment flow

Projects

FluxLLM — AI Gateway | *FastAPI, PostgreSQL, Redis, React, Docker, vLLM*

- Self-hosted LLM gateway unifying Groq, Gemini, Ollama, and vLLM behind one OpenAI-compatible API
- Adaptive EMA routing scores models on live cost and latency; per-provider circuit breakers with automatic fallback and SHA-256 response caching keep it resilient and cheap

Work Memory System | *Python, FastAPI, RAG, PostgreSQL/pgvector*

- Document-grounded RAG over structured work data with PDF/DOCX ingestion and Gemini embeddings
- Performed data preprocessing, feature extraction, and evaluation of retrieval quality metrics to improve response accuracy.
- Intent-aware reranking and deduplication handle temporal and experience-based queries with grounded, cited answers

King of the Pot | *Solidity, Foundry, Viem, Redis, PostgreSQL, Socket.io*

- On-chain PvP jackpot game on Base where players overtake a USDC pot to remain the last "King" when the timer expires
- Single-contract room system (ReentrancyGuard, CEI pattern, full test coverage) with a Viem event indexer feeding Redis/Postgres and real-time WebSocket updates

Core Technologies

Programming & Data: Python, SQL, NumPy, Pandas, PostgreSQL

Machine Learning & Analytics: Machine Learning, Supervised Learning, Unsupervised Learning, Classification, Regression, Decision Trees, Random Forest, Gradient Boosting, Feature Engineering, EDA, Statistical Analysis, Model Evaluation (ROC-AUC, Precision, Recall, F1), Predictive Modeling, Time Series Forecasting, Scikit-learn

AI & LLM Systems: PyTorch, HuggingFace Transformers, RAG Systems, LangChain, OpenAI GPT, Google Gemini, Claude, Groq, Pinecone

Backend & Systems: FastAPI, Node.js, Golang, REST APIs, Redis, Docker, Nginx

Cloud: Alibaba Cloud; familiar with AWS

Additional: TensorFlow, OpenCV, React, TypeScript

Certifications

- IBM – Supervised Machine Learning: Classification
- IBM – Unsupervised Machine Learning
- IBM – Exploratory Data Analysis for Machine Learning
- DeepLearning.AI – Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
- DeepLearning.AI – Natural Language Processing with Classification and Vector Spaces
- National Taiwan University – Operations Research (2): Optimization Algorithms

Additional

- Token2049 Singapore (Oct 2024) – Represented Tychi Labs at premier international blockchain & AI conference; demonstrated live AI-powered wallet features to global investors and ecosystem leaders